

ZEESHAN AHMED

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EDUCATION

IIIT Hyderabad

B.Tech. and M.S. in Computer Science and Engineering / GPA: 8.77

July 2019 – July 2026

Hyderabad, India

EXPERIENCE

FYDY (Fuzzy Dynamics Inc.)

Co-Founder and CTO

October 2025 – Present

Hyderabad, India

- Lead development and hill-climbing of OpenScientist's agent harness for long-horizon research, structuring environments around task sets, harnesses, and verifiers.
- Built infrastructure for parallel agent orchestration, experiment execution, steering, verification, and reusable trace capture across coding, mathematics, and science.
- Scaled the platform to support thousands of collaborating agents and coherent research trajectories spanning days to weeks.
- Co-lead research on adaptive test-time compute, out-of-distribution reasoning, and continual learning for scientific discovery and experimentation.

Microsoft

Software Engineer

July 2024 – August 2025

Delhi, India

- Developed a redesigned editor for the Microsoft Designer Android application, expanding AI-powered photo-editing workflows and user-facing editing capabilities.
- Implemented richer editing and creation experiences within the Android product, contributing to the evolution of the app's core editor.
- Contributed to Microsoft Create as part of the Microsoft Office Suite, building user-facing creative tooling across the broader product surface.

QunaSys

Software Engineering Intern

June 2022 – September 2023

Tokyo / Remote

- Maintained and extended QURI Parts, QunaSys's modular open-source Python library suite for portable quantum algorithms.
- Added Qiskit Primitive and Runtime integration, quantum measurement and classical-bit support, ideal samplers, hardware connectivity access, and sampling fixes.
- Standardized framework interoperability across 5 open-source quantum SDKs, including Google Cirq.
- Implemented low-memory noise modeling for quantum tensor-network simulation.
- Developed hardware-tailored diagonalization using group-theoretic structure, reducing VQE runtime by 2 orders of magnitude.

Summer of Bitcoin

Intern at Revault

May 2022 – August 2022

Remote

- Extended a user-facing Bitcoin system with Child Pays for Parent (CPFP) transaction support across the CLI and Iced Rust GUI.

IIIT Hyderabad

Lead Teaching Assistant

Spring 2021 – 2024

Hyderabad, India

- Led the Automata Theory teaching team during Spring 2021, 2022, and 2023, managing course logistics for 250 students.
- Teaching Assistant for Quantum Algorithms in 2024.

RESEARCH & PUBLICATIONS

M.S. Thesis: Early fault-tolerant quantum algorithms for Hamiltonian simulation: benchmarking and applications  June 2026

Advisor: Prof. Shantanav Chakraborty

IIIT Hyderabad

- Benchmarked Single-Ancilla LCU, first- and second-order Trotterization, and qDRIFT across sparse Ising chains and dense quantum-chemistry Hamiltonians, quantifying resource trade-offs over precision, evolution time, and system size.
- Applied Hamiltonian-simulation methods to ground-state magnetization across a quantum phase transition, identifying structure- and precision-dependent operating regimes.
- Developed collision-model algorithms for Markovian and non-Markovian open-system dynamics using a single environment qubit and minimal ancilla overhead.

Selected publications

- [1] K. Garg, **Z. Ahmed**, S. Mitra, and S. Chakraborty. *Simulating quantum collision models with Hamiltonian simulations using early fault-tolerant quantum computers*. *Physical Review A* 112, 022425 (2025). 
- [2] K. Garg, **Z. Ahmed**, and A. Thomasen. *Entanglement determination with a deep multiscale entanglement renormalization ansatz*. *Physical Review A* 111, 042411 (2025). 
- [3] **Z. Ahmed**, A. Chaudhuri, K. S. S. Grover, A. Rao, K. Garg, and P. Malhotra. *Classifying CELESTE as NP Complete*. *CST 2022*. 

PROJECTS

PinguDB: Relational Database Management System 

C++, CMake, Systems

- Developed an RDBMS with an SQL parser and multi-table operations including JOIN and SORT with $O(n)$ overhead.
- Engineered explicit paging and buffer management to support multiplication of matrices of at least 100 MB.

Nano-pass Compiler 

Racket

- Constructed a nano-pass compiler in Racket for a subset of the language.
- Added vectors and garbage collection; implemented loops through $O(1)$ -space tail recursion.

Quantum Molecule Simulator 

Python, Qiskit

- Constructed a circuit-based simulator for molecular Hamiltonians represented in an extended Pauli basis using product formulas.
- Used multi-pass transpilation to reduce circuit length by 50%, with a precision trade-off for shorter circuits.

Fragments: 2D Puzzle Platformer 

C#, Unity

- Built a multi-level puzzle platformer around detachable, independently controlled body parts that players reassemble to unlock movement and traversal mechanics.
- Implemented gameplay systems for enemies, combat, inventory, dialogue, HUD, camera tracking, and scene progression.

ACHIEVEMENTS

- **PauliZee**: Winner of the QHack '22 Quantum Chemistry challenge organized by **XanaduAI**.
- **Christine**: Overall winner at Kent Hack Enough 2020.
- **Aristos**: Runner-up at **Megathon 2022**, IIIT Hyderabad.
- Named to the **Dean's Merit List** for consistent academic performance.

REFERENCES

Shantanav Chakraborty

Assistant Professor, IIIT Hyderabad

Andreas Thomasen

Quantum Engineer, QunaSys

Tennin Yan

CEO and Co-founder, QunaSys

Sourabh Raheja

Principal Engineering Manager, Microsoft